Postdoctoral Associate

Wild bee heat wave resilience

(Job Code: 9546)



iBUG LAB
INSECT BIODIVERSITY UNDER GLOBAL-CHANGE

The iBUG lab (in collaboration with the Williams Lab at UC Davis and the Winfree Lab at Rutgers) is looking for a postdoctoral researcher to join us on a new, NSF-sponsored project investigating the resilience of wild bees to heat waves. This position offers up to 2 years of guaranteed funding with possibility of extending to a third given successful performance. Our project bridges the study of individuals, populations and communities and leverages cutting-edge technology to better understand organism responses to the environment. First, the project measures behavior, reproduction and survival of individual bees across life stages, in response to simulated heat waves, focusing on *Osmia lignaria*. Second, the project tracks nesting *Osmia* in the wild using automated remote cameras to determine whether bees within a population choose cooler nest locations under heat wave conditions. Third, the project uses AI-assisted camera traps to measure the activity of whole native bee communities before, during, and after naturally occurring heat waves. The postdoc will lead the first objective of the project, including the development of experimental protocols, data collection, analysis, and publishing results. Additional opportunities will exist to work on the second objective as well. All experimental and field work will be conducted at UC Davis/in California (housing provided), but the postdoc will be based at the University of Minnesota for all other periods of the project.

Responsibilities (non-exhaustive list)

- *Research and coordination (75%):* Experimental design, development of protocols, insect/plant rearing, carrying out experimental heat wave simulations, collecting and analyzing data.
- *Publication preparation (20%):* Preparing and leading publications of experimental results in collaboration with project PIs and other participants.
- *Mentorship (5%):* Mentor project technicians and undergraduate and graduate students in scientific practices, writing, and project management.

Minimum Qualifications:

- PhD in Ecology, Entomology, or other related ecological disciplines (those with scheduled defense date in Winter 2026 will be considered).
- Experience managing field work in exposed, high summer temperatures (≥ 95°F).
- Valid drivers' license.
- Experience with *Osmia* in experimental contexts in addition to growing and maintaining plants in greenhouse conditions.
- Experience with R programming, including basic to intermediate analysis of experimental and observational research designs.

Preferred Qualifications:

- Demonstrated publication record.
- Familiarity with field identification of bee species.
- Advanced statistical techniques such as generalized linear mixed models (GLMMs) and generalized additive models (GAMs).
- Interest in designing and deploying with cameras to monitor *Osmia* nest blocks and working with the associated data processing pipelines

Project Duration: March 1, 2027 - Feb. 29, 2028

Salary and Conditions:
Salary: Full-time salary (minimum $62,232 per year) and benefits included. Salary and benefits are consistent with University of Minnesota policy and commensurate with applicant experience.
Start date: Priority will be given to applicants who can start by March 1, 2027.

The initial appointment is for one year with reappointment for a second year dependent on positive performance during year 1. Renewal beyond 2 years is possible depending on funding for the project or the candidate's ability to obtain continued funding from other sources (e.g., USDA Postdoctoral Fellowship, Smith Fellowship, or other research grants in collaboration with PIs).

Application Materials and Deadline:
Applicants should submit a brief cover letter, CV, a 1-page summary of research interests, 1-3 representative publications (or formatted manuscripts/thesis chapters), and the names and contact details of at least two references by December 11, 2026 for full consideration.

We will begin reviewing applications December 11, 2026. Co-PI Williams will be attending the Entomological Society of America meetings in Columbus and is happy to discuss the project and position with prospective applicants.Please also reach out to Jeremy if you have any additional questions.